The Buttons and How an Install Works

What each instmon button sends, the checks that stand in front of each action, how instmon installs itself with `--install`, and how it installs any other tool from the library with that tool's own `--install`.

BUTTONS · --INSTALL · LIBRARY INSTALL · CONFIRMATIONS

SUMMARY · READ-ONLY REVIEW

Every action the page can send

Each button posts one action name to `/api/action`. The server looks the name up in one table and calls its handler. The table is the complete list of what the page can ask instmon to do.

GROUP	ACTIONS	WHAT THEY DO
Components	install, uninstall, start, stop, delete, lib_lock	Install or remove a tool from its library file; start or stop its service; delete a library file; lock or unlock a library file so it cannot be overwritten
Config	install_config, save_alternate	Copy a library config file into the node's config folder; save a live config as an alternate
Scripts	run_script	Run a script from the scripts library
Updates	gh_check, gh_update, gh_channel, build_pref, full_update, fu_stop, fu_dismiss, app_fetch, app_delete	Check GitHub, download one file, choose the channel and build, run or stop a Full Update, fetch or delete the app
Helpers	install_confgetter, cg_run, cg_uninstall, cg_delete; install_confputter, cp_uninstall, cp_preview, cp_apply, cp_undo, cp_delete_edited; rpiclone_install	Install and run Confgetter and Confputter, preview and apply a Confputter change and undo it, install rpi-clone
System	quiet, restore, comms_restart, reboot, shutdown	Quiet System pause and restore; restart the radio services; reboot or power off the node

Some actions are refused until the request says `confirm: true`: install, run_script, rpiclone_install, the install and uninstall of Confgetter and Confputter, gh_update, full_update, fu_stop, and the switch to the Testing channel. Others do not ask for it in the server (for example uninstall of a tool, install_config, delete, start, stop, comms_restart, reboot and shutdown); the page asks the operator before it sends some of those.

What stops an action from running

- **Full Update.** While a Full Update is running, every action except `fu_stop` and `fu_dismiss` is refused with 409.
- **Recording.** While Node Recorder is recording (instmon asks its status port, 127.0.0.1:8995, at most once every 2 seconds), 22 settings, install and service actions are refused with 409: install, uninstall, start, stop, install_config, run_script, rpiclone_install, save_alternate, comms_restart, reboot, shutdown, gh_update, quiet, restore, full_update, and the Confgetter and Confputter install, run, uninstall, apply and undo actions. The message says the lock lasts until the recording ends, so it is not interrupted.
- **Confirmation.** The actions listed in Part 1 need `confirm: true`, or they are refused with 400.
- **Disk-image job.** An install, an uninstall, a script run, a rpi-clone install, a reboot or a shutdown is refused with 409 while a disk-image job is running (Part 4 of this series).
- **One install at a time per tool.** A second install for the same tool is refused while the first is running (409). The same applies to a config install onto the same file.
- **One run at a time per script.** A script that is already running cannot be started again (409).
- **No self-stop.** instmon refuses to stop itself from its own panel. It can restart itself only by the scheduled restart at the end of its own install (Part 3.1).
- **Library lock.** A locked library file cannot be overwritten by an upload, an update, or an editor save (Part 2 of this series).
- **Config names.** A library config is installed only if its name maps to a known node config. A Node Talker config (nodetalker or nodelisten) must contain its own section, or it is refused.

The `--install` option for instmon

3.1 What `--install` does

The program's argument parser accepts `--install` and `--uninstall`. With `--install` it runs the install routine, which needs root (it exits with an error otherwise). The routine:

1. Copies the running file into the install folder `/usr/local/lib/asl_dvs/instmon`, under its own name, with mode 755.
2. Replaces the link at `/usr/local/bin/instmon.py` with one that points at that copy. A plain file at that path is replaced, and its compiled cache files are removed.
3. Removes the other Python copies in the install folder, so only the new one is left.
4. Writes the launcher, and the service file `/etc/systemd/system/instmon.service`. The service is Type=notify, runs the shared launcher with the installed file, and restarts always after 3 seconds, with a watchdog every 60 seconds.
5. Opens instmon's port (8990) in the firewall, through the shared firewall helper.
6. Reloads systemd and enables `instmon.service`.
7. Fetches the manuals from the repository (the docs fetch, Part 1 of this series).
8. Schedules a restart of `instmon.service` two seconds later, through a one-shot systemd timer. The restart is scheduled and not run inline, so the install can finish before its own process is stopped.

3.2 What `--uninstall` does

It also needs root. It disables and stops `instmon.service`, removes the service file, reloads systemd, removes the link at `/usr/local/bin/instmon.py` and the installed copy it pointed to, removes its compiled cache and empty install folders, removes the launcher if no other service file names it, and closes the firewall port. The library, config and install folders under the library path are left in place. The panel cannot start this on itself: instmon refuses to uninstall itself (400).

The Install button on a tool

The Install button installs one tool from its library file. instmon does not know how each tool installs itself. It hands the work to the tool's own `--install`, and watches the log.

4.1 Before anything runs

1. **Confirmation** (Part 2).
2. **The file.** The file must be in the library folder for the tool's category.
3. **Self-install check.** The file must have a self-install routine. For a Python file, its text must have an `--install` argument and define `install_service`. For a shell file, it must contain an `--install)` case and a `do_install` function. A file without them is refused with "doesn't support self-install … Nothing was run."
4. **Syntax check.** A Python file is compiled, and a shell file is checked with `bash -n`. A syntax error stops the install before anything changes.
5. **One install per tool** (Part 2).

4.2 What the background run does

1. **Prepare the install.** Any symlink on the install path is replaced by a real copy of its target, so the new install writes a real file and does not change the old one through the link. Other unit names that the new file does not use are retired: the unit is disabled and removed, along with its unit paths.
2. **Run the file's own `--install`** with the Python or shell interpreter that suits it (`python3` or `bash`), in instmon's own environment plus `PYTHONUNBUFFERED=1`. The button run is a child of the instmon process. (The Full Update runs its commands as a separate systemd unit.)
3. **Stream the output** to instmon's log line by line.
4. **Wait up to 600 seconds** (INSTMON_SCRIPT_TIMEOUT_SEC). A run that takes longer is killed, and the log says so.
5. **On success** the log records it, the quiet-mode entry for the tool is cleared, and, for every tool except instmon, a watch starts that checks the tool stays up for 60 seconds. If it does not, the log says to check it, and points to the previous version in the library.
6. **On failure** the log records the exit code. Nothing is rolled back.

The run does not roll back. A button install that fails leaves the tool as the failed install left it. The Full Update (Part 1 of this series) is the path that rolls back.

Uninstall, start, stop and delete

- **Uninstall** runs the tool's own `--uninstall`. The installed file must exist and must have an uninstall routine, or the uninstall is refused (400). instmon cannot uninstall itself. Before it runs, instmon saves a copy of the installed file into the library (unless an identical copy is already there), and it refuses to go on if that copy cannot be made. It is refused while a disk-image job is running, and while another install or uninstall of the same tool is in progress (409).
- **Start and stop** act on the tool's service through systemctl. "Start" is a restart. instmon refuses both for itself. After the command, the handler waits half a second and reports the service state it finds.
- **Delete** removes a file from the library. The category must be known (400), the file must exist (404), and a locked file is refused (409). Delete asks for no confirmation in the server.
- **Lock** marks a library file as locked, or clears the mark. A locked file is protected from upload, update and editor overwrite (Part 2 of this series).

Power, radio services and Quiet System

- **Reboot and shutdown** are scheduled, not immediate: a one-shot systemd timer runs the command after about 2 seconds (INSTMON_REBOOT_SHUTDOWN_DELAY_SEC). The page is told the node will be unreachable until it comes back up. A shutdown needs physical access (or a smart PDU) to power it back on. Both are refused while a disk-image job runs.
- **Restart radio services** (comms_restart) stops Asterisk, Analog_Bridge, MMDVM_Bridge, STFU and Allmon3 in that order, pauses 3 seconds, and starts them in the same order. It reports each one as ok or FAILED (state after start), with 207 and an error level when any failed.
- **Quiet System** pauses the services that compete for the SD card (the page lists SysMon, the Dashboard, 44helper, Node Talker, Node Recorder, Pibrary and any watchdog timer, and says the radio, Asterisk, the bridges, Allmon3, SSH and wifimon stay up), and **Restore** brings them back. Quiet is refused while a disk-image job runs. It restores itself after 30 minutes (INSTMON_QUIET_AUTO_RESTORE_SEC). When a tool's install succeeds, its quiet entry is cleared (Part 4.2).

What the install path shows

7.1 Each tool installs itself

instmon never copies a tool into place by itself. Each tool's own `--install` does the work, so the book should describe the install as the tool's, with instmon as the caller.

7.2 A button install does not roll back

The Full Update rolls back when a tool does not come back. The Install button does not. Operators should be told to use the Full Update, or to keep the previous version in the library, if an install may fail.

7.3 Checks catch the obvious faults

Confirmation, the self-install check, the syntax check and the busy checks stop bad requests before anything changes. They do not check that the new tool works. That is the health watch's job after the install, and it runs only for a minute.

7.4 Not checked

- The individual install routine inside each tool (covered by each tool's own summary).
- The exact cleanup of the install folder for tools other than instmon.
- The Confgetter and Confputter helper actions in full (only their names and checks were read).